

FREUID Challenge 2026 — Technical Report

1. Introduction

We address identity-document fraud detection as a binary image-classification problem: each document image is assigned a continuous fraud score (higher = more likely fraudulent), evaluated with the challenge’s tail-focused DET metric (APCER at low BPCER). The task exhibits a strong domain shift between the training data and the (public/private) test data — including print-and-capture recaptures and document types under-represented in training — which drives our design choices toward strong local-artifact modelling at inference time.

2. Method

Backbone. `maxvit_base_tf_512.in21k_ft_in1k` (timm), a hybrid conv + window/grid-attention model at 512×512 . In an earlier fold-0 sweep of 30+ timm backbones, MaxViT @ 512 clearly dominated (mid-size hybrid + high resolution beat larger pure-conv / pure-ViT models that overfit the digital domain), which is why it is our final backbone.

Training data. All competition training images with the 5 stratified folds merged (no held-out validation — in-domain OOF is saturated and does not discriminate models), plus **self-generated pseudo-labels**: confident predictions on the public test images (score $< 0.05 \rightarrow$ bona-fide, $> 0.95 \rightarrow$ fraud) added to the training set (semi-supervised). **No external datasets are used.**

Objective / optimisation. BCEWithLogitsLoss on a single logit; AdamW (lr $1e-4$, weight-decay $1e-5$); cosine schedule with 1-epoch warmup; bf16 autocast; channels-last. Image size 512, batch size 10, 12 epochs.

Augmentation. “V1”: horizontal-flip, vertical-flip, rotate($\pm 20^\circ$), then resize the whole document to 512. (Crop-based *training* was tried and hurt; crops help only at *test* time — see §5.)

Checkpoint ensemble. We save epochs 10, 11, and 12 and ensemble all three; averaging the three neighbouring epochs denoises the pseudo-label-trained model.

3. Inference

For every test image each of the 3 checkpoints is scored with **dense-crop test-time augmentation**:

- views = the native 512 resize + a 3×3 grid of crops taken from a 1.15× up-scaled image (9 crops), each shown in the 4 orientations of the D2 group (identity, h-flip, v-flip, 180° rotate) → **10 × 4 = 40 views**;
- the 40 view probabilities are averaged, then the 3 checkpoints are averaged → one fraud score per image.

Rationale: fraud cues (font/MRZ/splice/GenAI edits) are *local*; a full-image resize blurs them, whereas averaging many local crops preserves them. The inference is deterministic and needs no network (pretrained=False; weights from local checkpoints).

4. Results

Public leaderboard (AuDET, lower = better):

Configuration	Public
Single checkpoint (e12), dense-crop TTA	0.00178
3-checkpoint ensemble (e10+e11+e12), dense-crop TTA — final	0.00145

5. Reproducibility

- **Code / weights frozen** at the tagged commit (see the pinned Kaggle reply for the exact SHA); no model, architecture, or training-code changes after the freeze date.
- **Deterministic, network-free inference** via the provided Docker image, run with `--network none`; weights are baked into the image (Git LFS in the repo).
- **Data sources**: competition training set + self-generated pseudo-labels on the public test set. No external data, no runtime downloads.
- **Training entrypoints**: `train/` (`make_folds.py`, `pseudo_gen.py`, `train_full.py`) reproduce the checkpoints from the competition data; commands are documented in `train/README_TRAIN.md`.
- **Hardware**: single NVIDIA H100 (80 GB), CUDA 12.6, PyTorch 2.7.1, timm 1.0.27.
- **Measured runtime**: the Docker image (`--network none`, `bf16`, `channels-last`, `--bs 16` `--shm-size=8g` `--workers 8`) scores the full **7,821-image public set in ≈65 min** on one H100 — i.e. ≈0.5 s/image for the 40-view × 3-checkpoint dense-crop TTA (120 forward passes/image). The 134,997-image private set scales to **≈19 h on a single H100**; use several GPUs (one shard of /data each) or a larger `--bs` to cut wall-clock. The default `--bs 4` is memory-safe (<11 GB, no `--shm-size` needed) but slower; raise it on ≥24 GB GPUs.

6. Notes / limitations

The gains at the extreme low-FPR tail are small in absolute terms (a handful of tail images on the 5% public split), so ranking there is sensitive; the ensemble + dense-crop TTA is chosen for stability rather than chasing the noise floor.